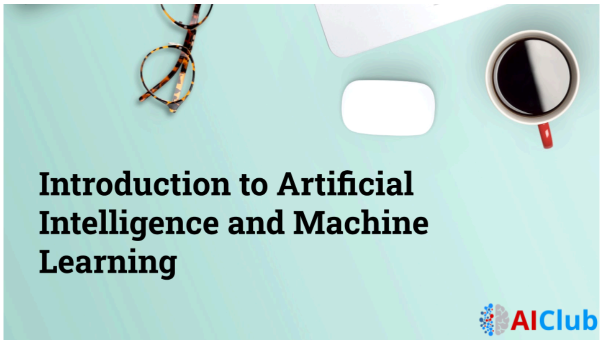
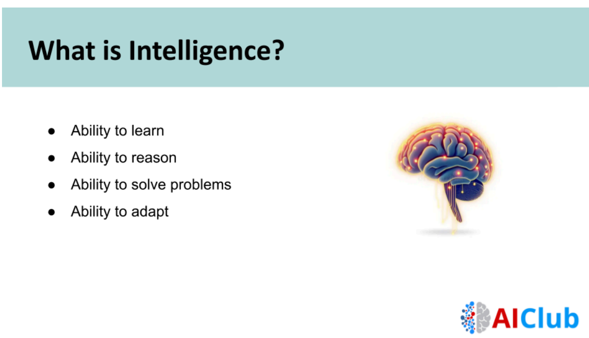
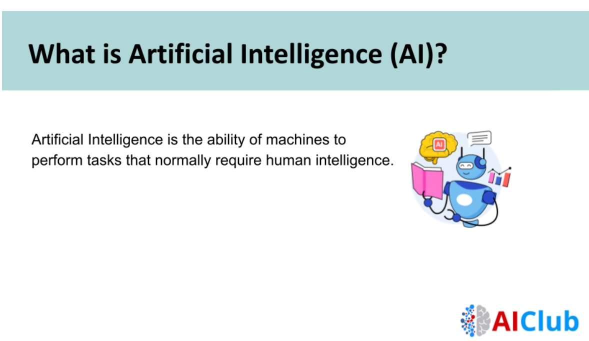
Introduction to AI - Teacher’s guide

	Hello everyone, welcome to AIClub. Today we will learn about Artificial Intelligence, also called AI, and Machine Learning, also called ML. These technologies are used in many tools you already use. By the end of this lesson, you will understand what AI and ML are and how they work. Before we talk about AI, let’s first understand what intelligence means.
	Intelligence means the ability to learn, reason, solve problems, and adapt to new situations. Humans use intelligence every day, like solving math problems, learning new skills, and making decisions. Now that we understand intelligence in humans, let’s see what Artificial Intelligence means.
	Artificial Intelligence is when machines can do tasks that normally require human intelligence. This means machines can learn, think, and make decisions like humans in some ways. Now let’s look at some examples of AI that you use in your daily life.

What is Artificial Intelligence (AI)?

Artificial Intelligence is the ability of machines to perform tasks that normally require human intelligence.

Examples:

- Face unlock
- Google Maps
- Chatbots
- Recommendation systems

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Some examples of AI include face unlock on phones, Google Maps, chatbots, and recommendation systems like YouTube or Netflix. These systems learn from data and help make your experience easier and faster.

Now let's look at a more formal definition of AI.

Formal Definition of AI

Artificial Intelligence is a branch of computer science that enables machines to mimic human intelligence such as learning, reasoning, and decision-making.

Key Words Highlighted:

✔ Learning
✔ Reasoning
✔ Decision-making

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Artificial Intelligence is a branch of computer science. It allows machines to mimic human intelligence, such as learning, reasoning, and decision-making. This means machines can improve their performance and make smarter decisions over time.

Now let's explore where AI is used in real life.

Real-Life Applications of AI

- Education (Personalized learning apps)
- Healthcare (Disease prediction)
- Transportation (Self-driving cars)
- Entertainment (Netflix/YouTube recommendations)

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AI is used in education, like personalized learning apps. It is used in healthcare to predict diseases. It is used in transportation, like self-driving cars. It is also used in entertainment, like Netflix and YouTube recommendations.

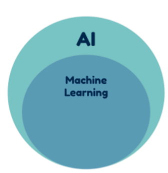
Now that we understand AI, let's learn about Machine Learning.

What is Machine Learning (ML)?

Machine Learning is a subset of AI that allows machines to learn from data and improve without being explicitly programmed.

Important Point:

- AI is a big concept
- ML is one way to achieve AI



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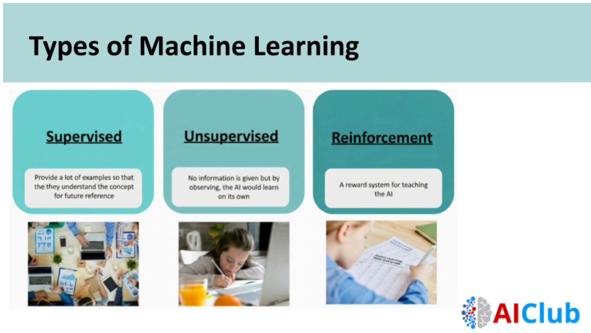
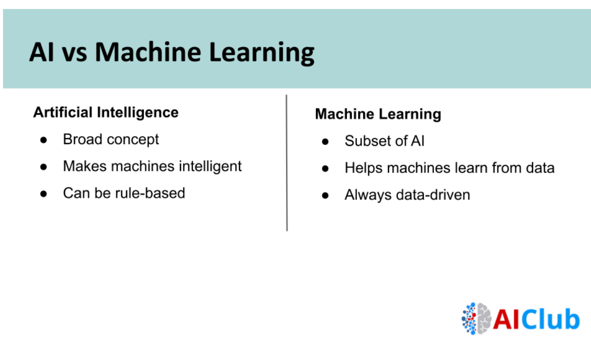
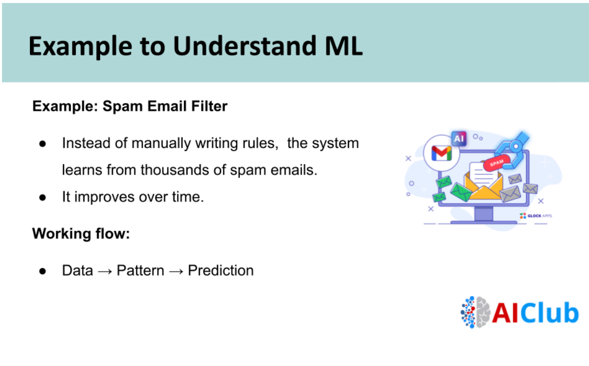
Machine Learning is a part or subset of AI. It allows machines to learn from data and improve without being directly programmed. AI is a big concept, and Machine Learning is one way to achieve AI.

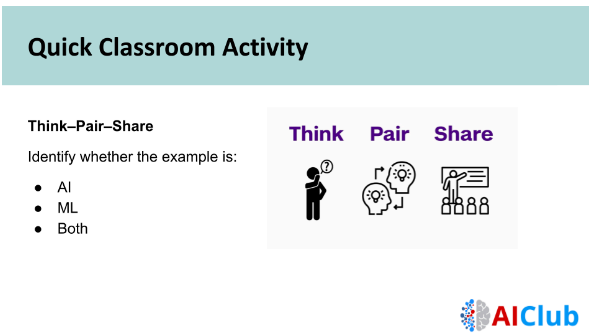
Now let's look at different types of Machine Learning.

	There are three main types of Machine Learning. The first is supervised learning. In this type, the model learns from labeled data, which means the correct answers are already provided. For example, it learns to identify cats and dogs from labeled pictures. The second is unsupervised learning. Here, the model does not have labels. It looks at the data and finds patterns on its own, such as grouping similar items. The third is reinforcement learning. In this type, the model learns through rewards and mistakes. For example, a game AI improves by learning which moves give better results. Now that we understand the types of Machine Learning, let's understand the difference between AI and Machine Learning.
	Artificial Intelligence, or AI, is a broad concept. It means making machines act smart, like solving problems or making decisions. Some AI systems use fixed rules. For example, a calculator follows set rules to give answers. That is rule-based AI. Machine Learning is a part of AI. Instead of only following rules, it learns from data. It studies many examples, finds patterns, and improves over time. The more data it gets, the better it performs. So, AI is a big idea, and Machine Learning is one way to achieve it using data. Now let's look at an example to understand Machine Learning better.



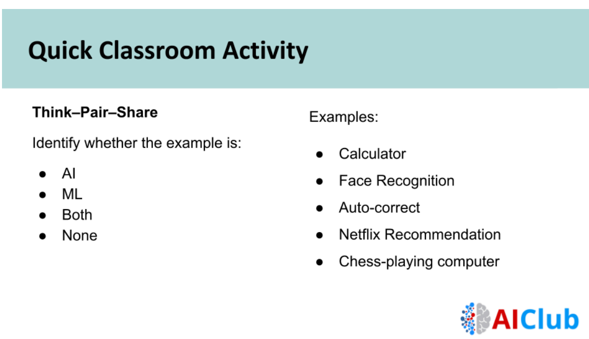
A spam filter is a good example of Machine Learning. Instead of writing rules manually, the system learns from thousands of spam emails. Over time, it improves and becomes better at detecting spam. The process follows this flow: data, pattern, and prediction.

Now let's do a short activity to check your understanding.



In this activity, you will think about examples and decide whether they use AI, Machine Learning, or both. First, think individually. Then discuss with a partner. Finally, everyone shares answers with the class.

Now let's look at some example technologies.



Examples include calculators, face recognition, auto-correct, Netflix recommendations, and chess-playing computers. Think about which ones use AI, Machine Learning, or both. This will help you connect what you learned to real-world technology.

Answers for teacher:
1) Calculator - None (Runs software in its internal chip)
2) Faces Recognition - AI & ML
3) Auto-correct - AI & ML
4) Netflix - AI
5) Chess play - ML

This brings us to the end of the lesson. You now have a basic understanding of Artificial Intelligence and Machine Learning.

	Thank you.